



Representing and Interpreting Data

· Mark Scheme

Statistics Chapter 3: Representing and Interpreting Data. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Histograms

- 1 (a) $30 \div 10 = 3$ B1; (b) $20 \div 5 = 4$ B1.
- 2 Class widths 10, 5, 5, 10, 20 M1; 1.2, 3.6, 5, 2, 0.5 A1 A1.
- 3 Frequency = fd $\times$ width M1; $4.5 \times 10 = 45$ A1; $2 \times 5 = 10$ A1.
- 4 (a) Area = 12 cm^2 represents 24 M1; 1 cm^2 represents 2 people A1. (b) Area = 10 cm^2 M1; frequency = 20 A1.
- 5(a) Area of 15–20 bar = 6 cm^2 for 30, so 5 customers per cm^2 M1 A1; 10–15 bar is 1 cm wide, area 5 cm^2 , frequency 25 A1.
- 5(b) Width 2 cm B1; area = $24 \div 5 = 4.8 \text{ cm}^2$, height 2.4 cm B1.
- 5(c) $\frac{3}{5} \times 25 + 30 + \frac{5}{10} \times 24$ M1 M1; $= 15 + 30 + 12 = 57$ A1.

Section B – Cumulative frequency and box plots

- 6 5, 17, 35, 45, 50 B1 B1.
- 7 Points plotted at upper class boundaries (10, 5), (20, 17), (30, 35), (40, 45), (50, 50), with (0, 0) B1; smooth curve B1; median at cf 25: about 24 years B1.
- 8 Q_1 : position 12.5, $10 + \frac{12.5-5}{12} \times 10 = 16.25$ M1 A1; Q_3 : position 37.5, $30 + \frac{37.5-35}{10} \times 10 = 32.5$ A1; IQR = 16.25 A1.
- 9 (a) IQR = 9, upper limit $27 + 13.5 = 40.5$ M1; $45 > 40.5$ A1. (b) Box 18 to 27 with median at 23, whiskers to 12 and 38 B1; 45 plotted as a separate cross B1.
- 10(a) 4, 14, 36, 54, 60 B1 B1.
- 10(b) Median: $40 + \frac{30-14}{22} \times 20 = 54.5$ M1 A1; $Q_1 = 40.9$, $Q_3 = 70$ M1; IQR = 29.1 A1.
- 10(c) 75% pass means the pass mark is the lower quartile M1; about 41 marks A1.

Section C – Comparing distributions

- 11 Class A scored higher on average (median 62 vs 58) B1; Class B's marks were more consistent (IQR 8 vs 15) B1.
- 12 Town P had more rain on average (mean 2.1 vs 1.4 mm) B1; Town P's rainfall was much more variable B1; the sd is larger than the mean, suggesting a few very wet days and many dry days (positive skew) B1.
- 13 Group Y were faster on average (median 18 vs 20 s) B1; Group Y were more consistent (IQR 4 vs 9 s, range 14 vs 27 s) B1; both answers in context B1.
- 14 House prices are usually skewed, with a few very expensive houses B1; these outliers distort the mean and standard deviation B1; the median and IQR are not affected by extreme values B1.
- 15(a) Leuchars: mean = 13.0 B1, sd = $\sqrt{\frac{5357.2}{31} - 12.987^2} = 2.04$ M1 A1; Hurn: mean = 16.8, sd = 1.99 A1.
- 15(b) Hurn was warmer on average (mean 16.8 vs 13.0°C) B1; the spread of temperatures was very similar (sd about 2.0°C at both) B1.
- 15(c) Expected, since the south of England is usually warmer than Scotland B1; only May 2015 was used, so it may not apply to other months or years B1.



Roar-level challenge

16 Frequencies 24, $5k$, 18 $M1$; $24 + 5k + 18 = 80 \Rightarrow k = 7.6$ $A1$; median is the 40th value, in $10 \leq x < 15$ (frequency 38) $M1$; $10 + \frac{40-24}{38} \times 5$ $M1$; $= 12.1$ $A1$.